Exercises from Ch. 2: Smooth Maps, Lee

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Last Updated: August 18, 2026

Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 2)
Prove Proposition 2.12 (smoothness of maps into product manifolds)

Proof. We state the proposition:

Proposition 0.1. (Proposition 2.12)
Suppose $M_1, \ldots, M_k$, and N are smooth manifolds with or without boundary, such that at most one of $M_1, \ldots, M_k$ has a nonempty boundary. For each i, let $\pi_i \colon M_1 \times \cdots \times M_k \to M_i$ denote the projection onto the M_i factor. A map $F \colon N \to M_1 \times \cdots \times M_k$ is smooth if and only if each of the component maps $F_i = \pi_i \circ F \colon N \to M_i$ is smooth.

Denote by $n_i = \dim M_i$ for each $1 \leq i \leq k$. We need a corollary.

Corollary 0.1. The map $\pi_i \colon M_1 \times \cdots M_k \to M_i$ is smooth.

Proof. Let $p = (p_1, \ldots, p_k) \in M_1 \times \cdots \times M_k$. For each i, choose a smooth chart (U_i, φ_i) containing $p_i = \pi_i(p)$. Then $(U_1 \times \cdots \times U_k, \varphi_1 \times \cdots \times \varphi_k)$ is a smooth chart containing $p = (p_1, \ldots, p_k)$. Denote $U = U_1 \times \cdots \times U_k$ and $\varphi = \varphi_1 \times \cdots \times \varphi_k$. The coordinate representation of π_i is given by

$$\varphi_i \circ \pi_i \circ \varphi^{-1} \colon \varphi(\underbrace{U \cap \pi_i^{-1}(U_i)}_{=U}) \to \varphi_i(U_i).$$

For any $x = (x_1, \ldots, x_k) \in \varphi(U) \subseteq \mathbb{R}^{n_1 + \cdots + n_k} \cong \mathbb{R}^{n_1} \times \cdots \times \mathbb{R}^{n_k}$ we have

$$(\varphi_i \circ \pi_i \circ \varphi^{-1})(x) = x_i.$$

This is smooth since it is a linear projection map from $\mathbb{R}^{n_1 + \cdots + n_k}$ to $\mathbb{R}^{n_i}$. Hence, $\pi_i \colon M_1 \times \cdots \times M_k \to M_i$ is smooth.

□

We return to the main proof.
Assume that $F \colon N \to M_1 \times \cdots \times M_k$ is smooth. Then $F_i = \pi_i \circ F \colon N \to M_i$ is smooth since it is the composition of smooth maps.
Conversely, assume that $F_i = \pi_i \circ F \colon N \to M_i$ is smooth for each $1 \leq i \leq k$. Let $p \in N$ and choose a chart (U, ψ) containing p. For each i, choose a smooth chart for M_i containing $F_i(p)$, say (V_i, φ_i). Let

$$V = V_1 \times \cdots \times V_k \quad \text{and} \quad \varphi = \varphi_1 \times \cdots \times \varphi_k.$$

We also have that

$$F(U) \subseteq V$$

by shrinking U if necessary. Then

$$\varphi \circ F \circ \psi^{-1} = (\varphi_1 \circ F_1 \circ \psi^{-1}, \dots, \varphi_k \circ F_k \circ \psi^{-1}).$$

Each component $\varphi_i \circ F_i \circ \psi^{-1}$ is smooth since F_i is smooth for every i. Therefore, $\varphi \circ F \circ \psi^{-1}$ is smooth as a map into $\mathbb{R}^{n_1+\cdot+n_k}$. Hence, $F\colon N \to M_1 \times \cdots \times M_k$ is smooth.

$\square$